

# AE-357: Radiation of [Ru(CO)(OH)<sub>2</sub>(PNP{tBu})] (AE-310) 365 nm and white, 1000 mM KOH

Date: 2024-10-16

Tags: Dihydroxo Radiation PNP  
P(tBu)N(py)P(tBu) [Ru(CO)(OH)<sub>2</sub>(PNP)]  
NMR AE 1H 31P

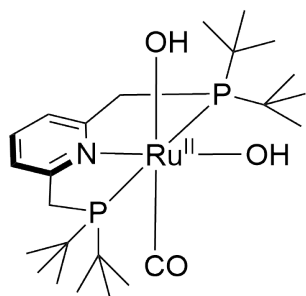
Category: Two-photon

Status: Done

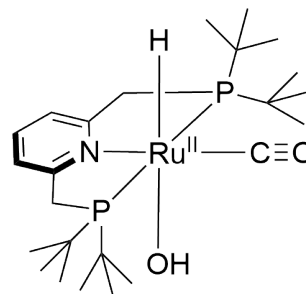
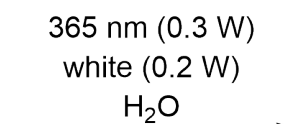
Created by: Alexander Eith

## Reaction scheme/sample structure

Radiation of Dihydroxo



Chemical Formula: C<sub>24</sub>H<sub>45</sub>NO<sub>3</sub>P<sub>2</sub>Ru  
Molecular Weight: 558,64552  
[Ru(CO)(OH)<sub>2</sub>(PNP<sup>tBu</sup>)]



Chemical Formula: C<sub>24</sub>H<sub>45</sub>NO<sub>2</sub>P<sub>2</sub>Ru  
Molecular Weight: 542,64652  
[Ru(CO)H(OH)(PNP<sup>tBu</sup>)]

## Literature/reference experiments

|                    |                                                                                                                                                                                                                                                                                                                          |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Literature         | Procedure after <a href="https://doi.org/10.1039/D1EE01053K">https://doi.org/10.1039/D1EE01053K</a>                                                                                                                                                                                                                      |
| Reproduction       | /                                                                                                                                                                                                                                                                                                                        |
| Related experiment | Experiment - AE-166: Radiation of [Ru(CO)(OH) <sub>2</sub> (PNP{tBu})], 92 % pure, 365 nm and white,<br>AE-246: Radiation of [Ru(CO)(OH) <sub>2</sub> (PNP{tBu})] (AE-238, 97 % pure) 365 nm and white<br>AE-348: Radiation of [Ru(CO)(OH) <sub>2</sub> (PNP{tBu})] (AE-310) 365 nm and white, water, longer irradiation |

## Reagents

| Name                                 | CAS/Experiment number                                                                                        | Amount [μmol] | Equivalents | Mass <sub>theo</sub> [mg] | Mass <sub>exp</sub> [mg] | Molar mass [g/mol] | Volume [ml] |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------|---------------|-------------|---------------------------|--------------------------|--------------------|-------------|
| [Ru(CO)(OH) <sub>2</sub> (PNP{tBu})] | AE-310: Synthesis of [Ru(CO)(OH) <sub>2</sub> (PNP <sup>tBu</sup> )] using Ag <sub>2</sub> O, 5 h (AE-310-2) | 4.55          | /           | /                         | 2.54                     | 558.65             | /           |
| KOH (1M)                             | AE-243: 1 M Stocksolution of KOH in water                                                                    | /             | /           | /                         | /                        | /                  | 0.90        |

# Irradiation Parameters

Power measured using [\[Power Meter\] 843-R-USB + 919P-020-12](#) unless specified otherwise.

|                     | Light Source Name                             | Wavelength [nm] | Power Setting [mW] |
|---------------------|-----------------------------------------------|-----------------|--------------------|
| First light source  | <a href="#">Light Source - LCS-0365-76-22</a> | 365 nm          | 300                |
| Second light source | <a href="#">Light Source - LCS-6500-65-22</a> | white           | 200                |

|                                   |                                               |
|-----------------------------------|-----------------------------------------------|
| Used beam combiner [Name or None] | <a href="#">Beam Combiner - LCS-BC25-0409</a> |
| Irradiation distance [cm]         | 9.5                                           |
| Thermostat temperature [°C]       | 25                                            |

## Procedure/observations

All steps were carried out using [\[Protocol\] Schlenk Technique](#). All steps were carried out in the dark, unless other specified.

| Date  | Time         | Step                                                                                                                                                                                                                                                              | Observations                        |
|-------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 15.10 |              | A 10 mL flask was prepared                                                                                                                                                                                                                                        |                                     |
|       | 10:10        | To the flask [Ru] was added under air                                                                                                                                                                                                                             |                                     |
|       |              | The flask was set under argon                                                                                                                                                                                                                                     |                                     |
|       | 15:00        | To the flask KOH (1M) was added                                                                                                                                                                                                                                   | A 1000 mM KOH solution was obtained |
|       | 15:05        | The obtained mixture was stirred at 320 rpm.                                                                                                                                                                                                                      | <a href="#">357 left.jpg</a>        |
|       | 15:35        | <a href="#">From the obtained solution an NMR sample was prepared according to Protocol - Preparation of NMR Sample using a D2O/H3PO4 capillary. (AE-357-1)</a> During the transfer the sample was filtered through a syringe filter (PA, pore diameter = 0.2 µm) | <a href="#">AE-357-1.jpg</a>        |
| 17.10 | 9:00         | The power of the LEDs 365 nm: 300 mW and white: 200 mW were adjusted according to <a href="#">[Protocol] Irradiation setup</a>                                                                                                                                    | <a href="#">power setup.jpg</a>     |
|       | 9:15         | The NMR-tube was placed in the setup according to <a href="#">[Protocol] Irradiation setup</a>                                                                                                                                                                    |                                     |
|       | 9:16 - 12:16 | The J.Young NMR tube was irradiated for 3:00 h                                                                                                                                                                                                                    | <a href="#">irrad setup.jpg</a>     |
|       | 15:30        | <a href="#">An NMR was measured (AE-357-2)</a>                                                                                                                                                                                                                    | <a href="#">AE-357-2.jpg</a>        |

## Analysis

| Date  | Time  | Sample name | Analysis Method         | Used Device    | Solvent | Raw Data                     | Processed Data                     | Comparative Data                                                                                                          | Interpretation                                                                                                                                              |
|-------|-------|-------------|-------------------------|----------------|---------|------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15.10 | 20:26 | AE-357-1    | 31P {1H}quant, 1Hzgespg | IAAC 400 MHz I | H2O     | <a href="#">AE-357-1.zip</a> | <a href="#">AE-357-1_10.nmrium</a> | /                                                                                                                         | New species formed with 25 % and shift of 65.0 ppm, otherwise clean spectrum                                                                                |
| 21.10 | 13:51 | AE-357-2    | 31P {1H}quant, 1Hzgespg | IAAC 400 MHz I | H2O     | <a href="#">AE-357-2.zip</a> | <a href="#">AE-357-2_10.nmrium</a> | Two-photon - AE-349: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, 50 mM KOH, shorter irradi (AE-349-2) | Full conversion. Hydride signal si observed at 91 ppm. Impurity at 57 ppm is not observed. Other impurities at: 81, 65, 59 and 55.5 ppm (all minor species) |

## Results

| Sample   | Yield (hydride species) [%] | Yield (dihydroxo) [%] | Data                                              |
|----------|-----------------------------|-----------------------|---------------------------------------------------|
| AE-357-2 | 42                          | 0                     | <a href="#">AE-357-2_Yield_Determination.xlsx</a> |

For comparison with other experiments see: [Two-photon - AE-371: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\] \(AE-310\) 365 nm and white, 0.1 mM KOH](#)

## Linked experiments

- AEI-022: Radiation of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNN})]$ , NO<sub>2</sub>-route, 365 nm and white LED
- AEI-026: Capillary production
- AEI-064: Synthesis of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$  with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag<sub>2</sub>O
- AEI-067: Radiation of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ , crude, 365 nm and white
- AEI-071: Radiation of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ , purer, 365 nm and white
- AE-105: Synthesis of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$  with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag<sub>2</sub>O
- AE-110: Radiation of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ , purified, 365 nm and white
- AE-119: Radiation of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ , purified, 52 mM KOH, 365 nm and white
- AE-140: Radiation of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ , purified, 365 nm and white, longer duration I
- AE-145: Synthesis of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$  with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag<sub>2</sub>O
- AE-158: Radiation of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ , purified, 365 nm and white
- KRA-092: Radiation of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ , 365 nm and white
- KRA-104: Degassing of Milli-Q water
- AE-238: Synthesis of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$  using Ag<sub>2</sub>O, 5 h
- AE-243: 1 M Stocksolution of KOH in water
- AE-246: Radiation of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$  (AE-238, 97 % pure) 365 nm and white
- AE-310: Synthesis of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$  using Ag<sub>2</sub>O, 5 h
- Two-photon - AE-348: Radiation of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$  (AE-310) 365 nm and white, water, longer irradiation
- Two-photon - AE-349: Radiation of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$  (AE-310) 365 nm and white, 50 mM KOH, shorter irradiation
- Two-photon - AE-371: Radiation of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$  (AE-310) 365 nm and white, 0.1 mM KOH

## Linked resources

Beam Combiner - [Beam Combiner LCS-BC25-0409 -1](#)

Light Source - [UHP LED 365 nm-1](#)

Light Source - [UHP LED 6500 K \(white\)-1](#)

Protocol - [Preparation of NMR Sample](#)

Protocol - [Schlenk Technique](#)

Protocol - [Irradiation setup](#)

## Attached files

AE-357-2\_Yield\_Determination.xlsx

sha256: 0c5744a31af703aed6bf2c551a1563a7b97afce8eea39067c90befb9b47a2321

AE-357-2\_10.nmrium

sha256: 8ae07392950132d6583673d42dca62575a7b047f59fe72eb7cd5a86f7d26ff08

AE-357-2.zip

sha256: a422c1c87b7a49d92410455f53525e4085bd6857e532a92e2acf0672bde69967

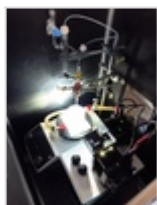
AE-357-2.jpg

sha256: 08883139c2b126abed140b270a19ee82311e9ebc24a78f5eae44e6b5b21d257c



irrad-setup.jpg

sha256: 924aa712744a9e6da6dada175aed06e28acbce3d18a73b07a29af768fc20a4c4



power-setup.jpg

sha256: 68e056b564d617fc961314440e9e7b452b3eaed78381a9215d0d00b72b5a036e



357-left.jpg

sha256: cae3301bce4dbae9e953d35c427997d51fe95f7b64a222193707722d0cb6c3b9



AE-357-1\_10.nmrium

sha256: c65a6c3bd627d1fd6fb6a9204a8992d88ab5dc75c47ebe3fec2dba523facf87

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AEI-06711.cdxml

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AE-357-1.jpg

sha256: 086625c9c86a405d044b194c56679caa470276302247705dd150953fa6f1414f



Unique eLabID: 20241016-efbdbf6b084191da002ef32c7220b60ffa318de  
Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1420>